Unit 1

Lesson Three

Ratio between two consecutive terms in the binomial expansion

In the expansion of $(a\,x + b\,y)^n$ according to the descending exponents of x

- **The ratio between two consecutive terms is :**

$$\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} \times \frac{b\,y}{a\,x} \longrightarrow \frac{n - r\text{ (the smaller)} + 1}{r\text{ (the smaller)}} \times \frac{\text{the second}}{\text{the first}}$$

- **The ratio between two coefficients of two consecutive terms is :**

$$\frac{\text{coefficient of } T_{r+1}}{\text{coefficient of } T_r} = \frac{n-r+1}{r} \times \frac{b}{a}$$

- **Note the difference between :**

$\frac{{}^nC_r}{{}^nC_{r-1}}$ and $\frac{T_{r+1}}{T_r}$:

while : $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} = \frac{n - r\text{ (the greater)} + 1}{r\text{ (the greater)}}$

For example : $\frac{{}^7C_5}{{}^7C_4} = \frac{7-5+1}{5} = \frac{3}{5}$

, but : $\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} \times \frac{\text{the second}}{\text{the first}}$

$$= \frac{n - r\text{ (the smaller)} + 1}{r\text{ (the smaller)}} \times \frac{\text{the second}}{\text{the first}}$$

For example :

In the expansion $(x + 5)^7$

we find $\frac{T_5}{T_4} = \frac{7-4+1}{4} \times \frac{5}{x} = \frac{5}{x}$

Example 1

In the expansion of $(3x - 4y)^{12}$ according to the descending exponents of x find each of :

1. $\frac{T_5}{T_4}$ **2.** $\frac{T_9}{T_{10}}$ **3.** $\frac{T_8}{T_6}$

Solution

$\because \frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} \times \frac{\text{the second}}{\text{the first}}$

1. $\frac{T_5}{T_4} = \frac{12-4+1}{4} \times \frac{-4y}{3x} = \frac{9}{4} \times \frac{-4y}{3x} = \frac{-3y}{x}$

2. $\because \frac{T_{10}}{T_9} = \frac{12-9+1}{9} \times \frac{-4y}{3x} = \frac{4}{9} \times \frac{-4y}{3x} = \frac{-16y}{27x}$ $\quad \therefore \frac{T_9}{T_{10}} = \frac{-27x}{16y}$

3. $\frac{T_8}{T_6} = \frac{T_8}{T_7} \times \frac{T_7}{T_6} = \left(\frac{12-7+1}{7} \times \frac{-4y}{3x}\right) \times \left(\frac{12-6+1}{6} \times \frac{-4y}{3x}\right) = \frac{6}{7} \times \frac{-4y}{3x} \times \frac{7}{6} \times \frac{-4y}{3x} = \frac{16y^2}{9x^2}$

Example 2

If the ratio between the third term and the fourth term in the expansion of $\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^6$ according to the descending exponents of x is 5 : 3 , find the value of x

Solution

$\because \frac{T_3}{T_4} = \frac{5}{3}$ $\quad \therefore \frac{T_4}{T_3} = \frac{3}{5}$ $\quad \therefore \frac{6-3+1}{3} \times \frac{\frac{1}{\sqrt{x}}}{\sqrt{x}} = \frac{3}{5}$

$\therefore \frac{4}{3} \times \frac{1}{x} = \frac{3}{5}$ $\quad \therefore 9x = 20$ $\quad \therefore x = \frac{20}{9}$

Example 3

If the second, third and the fourth terms of the expansion $(2x + y)^n$ according to the descending exponents of x are 40 , 20 , 5 respectively, find the values of each of n , x , y

Solution

$\because T_2 = 40 , T_3 = 20$ $\quad \therefore \frac{T_3}{T_2} = \frac{20}{40} = \frac{1}{2}$

$\therefore \frac{n-2+1}{2} \times \frac{y}{2x} = \frac{1}{2}$ $\quad \therefore \frac{n-1}{4} \times \frac{y}{x} = \frac{1}{2}$ by multiplying by 4

$\therefore (n-1) \times \frac{y}{x} = 2$ (1)

, $\because T_3 = 20 , T_4 = 5$ $\quad \therefore \frac{T_4}{T_3} = \frac{5}{20} = \frac{1}{4}$

$\therefore \frac{n-3+1}{3} \times \frac{y}{2x} = \frac{1}{4}$ $\quad \therefore \frac{n-2}{6} \times \frac{y}{x} = \frac{1}{4}$ by multiplying by 6

$\therefore (n-2) \times \frac{y}{x} = \frac{3}{2}$ (2)

By dividing (1) by (2) : $\therefore \frac{n-1}{n-2} = 2 \times \frac{2}{3} = \frac{4}{3}$

$\therefore 4n - 8 = 3n - 3$ $\therefore \boxed{n = 5}$

By substituting in (1) :

$\therefore (5-1) \times \frac{y}{x} = 2$ $\therefore 4y = 2x$ $\therefore 2y = x$

, $\because T_2 = 40$ $\therefore {}^nC_1 \, y^1 \, (2x)^{n-1} = 40$

By substituting as $n = 5$, $x = 2y$

$\therefore {}^5C_1 \, y \, (4y)^4 = 40$ $\therefore 5 \times y \times (4y)^4 = 40$ $\therefore 5 \times y \times 256 \, y^4 = 40$

$\therefore 1280 \, y^5 = 40$ $\therefore y^5 = \frac{40}{1280} = \frac{1}{32} = \left(\frac{1}{2}\right)^5$ $\therefore \boxed{y = \frac{1}{2}}$

, $\because x = 2y$ $\therefore \boxed{x = 1}$

Example 4

In the expansion of $(x + a)^n$ according to the descending exponents of x , if n is an odd number prove that : the ratio between the two middle terms respectively equals $\frac{x}{a}$

Solution

$\because$ n is an odd number

$\therefore$ The two middle terms are $T_{\frac{n+1}{2}}$, $T_{\frac{n+3}{2}}$

, $\because \frac{T_{\frac{n+3}{2}}}{T_{\frac{n+1}{2}}} = \frac{n - \left(\frac{n+1}{2}\right) + 1}{\frac{n+1}{2}} \times \frac{a}{x}$

$= \frac{2n - n - 1 + 2}{n+1} \times \frac{a}{x} = \frac{n+1}{n+1} \times \frac{a}{x} = \frac{a}{x}$

$\therefore$ The ratio between the two middle terms respectively $= \frac{T_{\frac{n+1}{2}}}{T_{\frac{n+3}{2}}} = \frac{x}{a}$

Finding the greatest term and the greatest coefficient in the binomial expansion :

In the expansion of $(a x + b y)^n$ according to the descending exponents of x , given the values of x and y , the greatest term numerically T_{r+1} is the term which is greater than or equal to the previous terms and greater than or equal to the next terms **i.e.** The term which satisfies both conditions

(1) $\frac{T_{r+1}}{T_r} \geq 1$ "greater than or equal to the previous terms"

i.e. $\frac{n-r+1}{r} \times \left|\frac{b y}{a x}\right| \geq 1$

(2) $\frac{T_{r+1}}{T_{r+2}} \geq 1$ "greater than or equal to the next terms"

$\therefore \frac{T_{r+2}}{T_{r+1}} \leq 1$ $\therefore \frac{n-(r+1)+1}{r+1} \times \left|\frac{b\,y}{a\,x}\right| \leq 1$

By using these two conditions , you can find the greatest term and the greatest coefficient in the expansion , the following example illustrates that.

Example 5

In the expansion $(2x - 3y)^{10}$ according to the descending exponents of x :

1. Find the numerical value of the greatest term at $x = 3$, $y = 2$

2. Find the numerical value of the greatest term at $x = 1$, $y = 3$

3. The greatest coefficient value in the expansion.

Solution

1. Let the greatest term be T_{r+1}

$\therefore \frac{T_{r+1}}{T_r} \geq 1$ "greater than or equal to the previous terms"

$\therefore \frac{n-r+1}{r} \times \left|\frac{b\,y}{a\,x}\right| \geq 1$ $\therefore \frac{10-r+1}{r} \times \left|\frac{-3 \times 2}{2 \times 3}\right| \geq 1$ $\therefore \frac{10-r+1}{r} \geq 1$

$\therefore 10 - r + 1 \geq r$ $\therefore 2r \leq 11$ $\therefore r \leq 5\frac{1}{2}$ (1)

, $\frac{T_{r+1}}{T_{r+2}} \geq 1$ "greater than or equal to the next terms"

$\therefore \frac{T_{r+2}}{T_{r+1}} \leq 1$ $\therefore \frac{10-(r+1)+1}{r+1} \times \left|\frac{b\,y}{a\,x}\right| \leq 1$ $\therefore \frac{10-r}{r+1} \times \left|\frac{-3 \times 2}{2 \times 3}\right| \leq 1$

$\therefore 10 - r \leq r + 1$ $\therefore 2r \geq 9$ $\therefore r \geq 4\frac{1}{2}$ (2)

From (1) , (2) :

$\therefore 4\frac{1}{2} \leq r \leq 5\frac{1}{2}$ $\therefore r = 5$

$\therefore T_6$ is numerically the greatest term in the expansion of $(2x - 3y)^{10}$ at $x = 3$, $y = 2$

, $T_6 = {}^{10}C_5 \left|(-3 \times 2)^5 (2 \times 3)^5\right| = 252 \times 6^{10}$

2. Let the greatest term be T_{r+1}

$\therefore \frac{T_{r+1}}{T_r} \geq 1$ "greater than or equal to the previous terms"

$\therefore \frac{n-r+1}{r} \times \left|\frac{b\,y}{a\,x}\right| \geq 1$ $\therefore \frac{10-r+1}{r} \times \left|\frac{-3\times 3}{2\times 1}\right| \geq 1$ $\therefore \frac{11-r}{r} \geq \frac{2}{9}$

$\therefore 99 - 9r \geq 2r$ $\therefore 11r \leq 99$ $\therefore r \leq 9$ (1)

, $\frac{T_{r+1}}{T_{r+2}} \geq 1$ "greater than or equal to the next terms"

$\therefore \frac{T_{r+2}}{T_{r+1}} \leq 1$ $\therefore \frac{10-(r+1)+1}{r+1} \times \left|\frac{-3\times 3}{2\times 1}\right| \leq 1$

$\therefore \frac{10-r}{r+1} \times \frac{9}{2} \leq 1$ $\therefore 90 - 9r \leq 2r + 2$

$\therefore 11r \geq 88$ $\therefore r \geq 8$ (2)

From (1) , (2) :

$\therefore 8 \leq r \leq 9$ $\therefore r = 8$ or 9

$\therefore$ There are two equal terms as the greatest term in the expansion

, $T_9 = T_{10} = {}^{10}C_9 \left|(-3\times 3)^9 (2\times 1)\right| = 20 \times 9^9$

3. To find the greatest coefficient value in the expansion , we find the value of the greatest term at $x = y = 1$

, let the greatest term be T_{r+1}

$\therefore \frac{T_{r+1}}{T_r} \geq 1$ "greater than or equal to the previous terms"

$\therefore \frac{n-r+1}{r} \times \left|\frac{-3}{2}\right| \geq 1$ $\therefore \frac{10-r+1}{r} \geq \frac{2}{3}$

$\therefore \frac{11-r}{r} \geq \frac{2}{3}$ $\therefore 33 - 3r \geq 2r$

$\therefore 5r \leq 33$ $\therefore r \leq 6\frac{3}{5}$ (1)

, $\frac{T_{r+1}}{T_{r+2}} \geq 1$ "greater than or equal to the next terms"

$\therefore \frac{T_{r+2}}{T_{r+1}} \leq 1$ $\therefore \frac{10-(r+1)+1}{(r+1)} \times \left|\frac{-3}{2}\right| \leq 1$

$\therefore \frac{10-r}{r+1} \times \frac{3}{2} \leq 1$ $\therefore 30 - 3r \leq 2r + 2$

$\therefore 5r \geq 28$ $\therefore r \geq 5\frac{3}{5}$ (2)

From (1) , (2) :

$\therefore 5\frac{3}{5} \leq r \leq 6\frac{3}{5}$ $\therefore r = 6$

$\therefore T_7$ at $x = y = 1$ is the greatest coefficient in the expansion

$\therefore T_7 = {}^{10}C_6 \left|(-3)^6 (2)^4\right| = 2449440$

Remarks

In the expansion of $(1 + X)^n$:

1. If n is an even number , then the greatest coefficient is the coefficient of the middle term $= {}^nC_{\frac{n}{2}}$
2. If n is an odd number , then the coefficients of the two middle terms are equal and each is the greatest coefficient in the expansion $= {}^nC_{\frac{n-1}{2}}$ or ${}^nC_{\frac{n+1}{2}}$

Example 6

In the expansion $(X+3)^{10}$ according to the descending exponents of X , find the values of X that make T_7 the greatest term numerically

Solution

$\because T_7$ is the greatest term

$\therefore \frac{T_7}{T_6} \geq 1$ $\therefore \frac{10-6+1}{6} \times |\frac{3}{X}| \geq 1$

$\therefore \frac{5}{6} \times 3 \geq |X|$ $\therefore |X| \leq \frac{5}{2}$

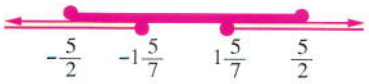
$\therefore X \in [\frac{-5}{2} , \frac{5}{2}]$ (1)

, $\frac{T_7}{T_8} \geq 1$ $\therefore \frac{T_8}{T_7} \leq 1$

$\therefore \frac{10-7+1}{7} \times |\frac{3}{X}| \leq 1$ $\therefore \frac{4}{7} \times 3 \leq |X|$ $\therefore |X| \geq 1\frac{5}{7}$

$\therefore X \geq 1\frac{5}{7}$ or $X \leq -1\frac{5}{7}$ (2)

From (1) , (2) :



$\therefore X$ that makes T_7 the greatest term $\in [\frac{-5}{2} , -1\frac{5}{7}] \cup [1\frac{5}{7} , \frac{5}{2}]$